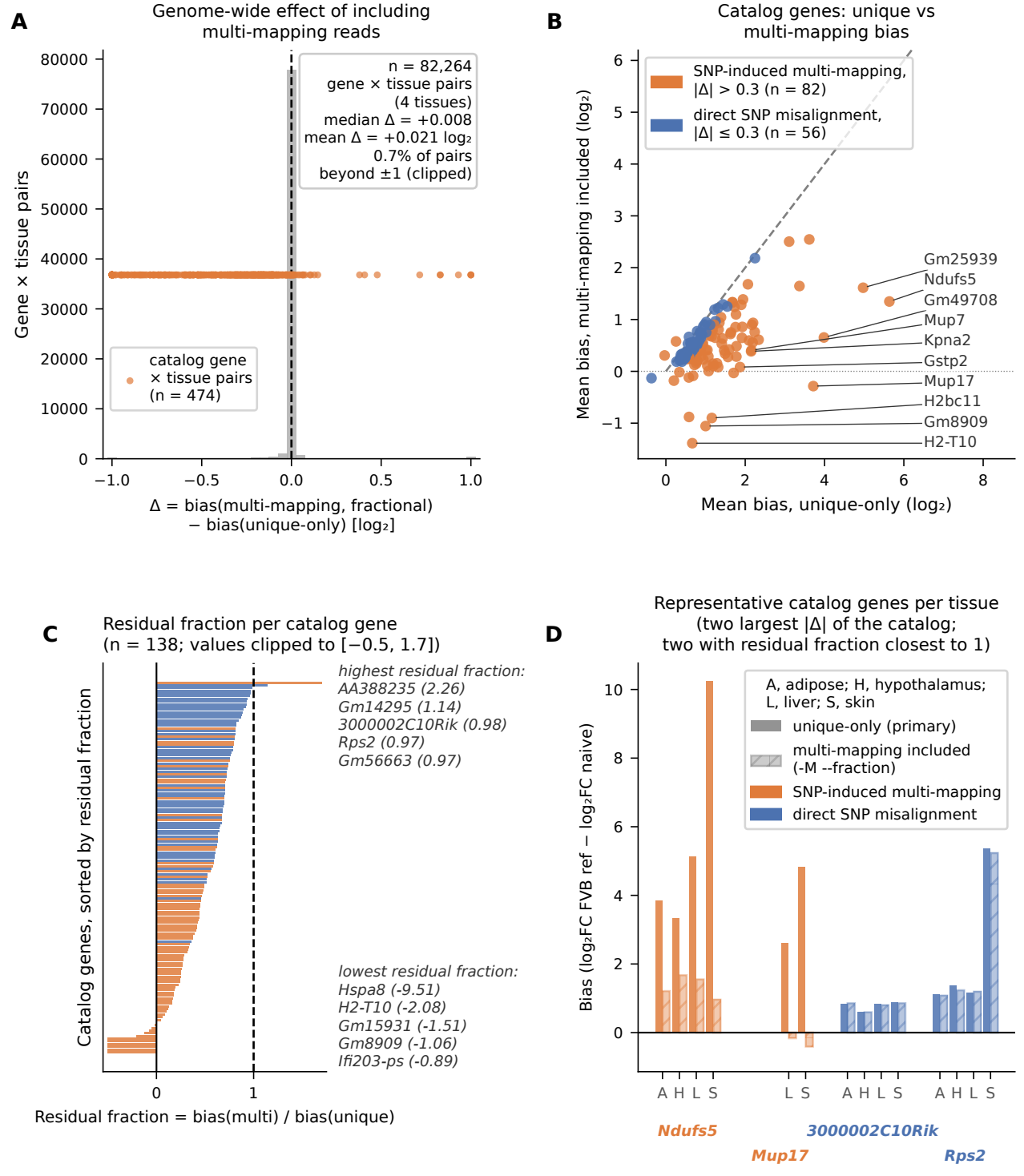

A
Genome-wide effect of including multi-mapping reads
n = 82,264 gene × tissue pairs (4 tissues)
median Δ = +0.008
mean Δ = +0.021 log₂
0.7% of pairs beyond ±1 (clipped)
catalog gene × tissue pairs (n = 474)
Gene × tissue pairs
Δ = bias(multi-mapping, fractional) − bias(unique-only) [log₂]
B
Catalog genes: unique vs multi-mapping bias
SNP-induced multi-mapping, |Δ| > 0.3 (n = 82)
direct SNP misalignment, |Δ| ≤ 0.3 (n = 56)
Gm25939
Ndufs5
Gm49708
Mup7
Kpna2
Gstp2
Mup17
H2bc11
Gm8909
H2-T10
Mean bias, multi-mapping included (log₂)
Mean bias, unique-only (log₂)
C
Residual fraction per catalog gene (n = 138; values clipped to [−0.5, 1.7])
highest residual fraction:
AA388235 (2.26)
Gm14295 (1.14)
3000002C10Rik (0.98)
Rps2 (0.97)
Gm56663 (0.97)
lowest residual fraction:
Hspa8 (-9.51)
H2-T10 (-2.08)
Gm15931 (-1.51)
Gm8909 (-1.06)
Ifi203-ps (-0.89)
Catalog genes, sorted by residual fraction
Residual fraction = bias(multi) / bias(unique)
D
Representative catalog genes per tissue (two largest |Δ| of the catalog; two with residual fraction closest to 1)
A, adipose; H, hypothalamus; L, liver; S, skin
unique-only (primary)
multi-mapping included (-M --fraction)
SNP-induced multi-mapping
direct SNP misalignment
Bias (log₂FC FVB ref − log₂FC naive)
A H L S
Ndufs5
Mup17
L S
3000002C10Rik
Rps2